

SDK Polyolefin Expansion Project		INTERLOCK LOGIC DIAGRAM & CAUSE / EFFECT MATRIX			DOC NO: TJC-LLD-IL-KC-4501 WORK NO: BA-0256 REV: 3	
LOGIC No: SEQ-4501	DESCRIPTION: RECYCLE GAS COMPRESSOR (KC-4501) SHUTDOWN LOGIC			SIL: SIL 2	TAG: KC-4501 FLOC: TJC-LLD-4500-01	

CAUSE & EFFECT MATRIX

ID	INITIATOR / CAUSE (INPUT)	TAG	SET POINT	VOTE	EFF-1 TRIP COMPRESSOR MOTOR KC-4501	EFF-2 OPEN ANTI-SURGE RECYCLE FV-4502	EFF-3 CLOSE SUCTION XV-4501	EFF-4 MAINTAIN SEAL GAS N2	EFF-5 ANNUNCIATE ALARM ON DCS
T1	Lube oil pressure LOW-LOW	PSLL-4504	< 1.5 barg	2oo3	X	X	X		X
T2	Discharge temperature HIGH-HIGH	TSHH-4503	> 140 degC	1oo1	X	X			X
T3	Discharge pressure HIGH-HIGH	PSHH-4502	> 14 barg	1oo2	X	X			X
T4	Crosshead vibration HIGH-HIGH	VSHH-4505	> 11 mm/s	1oo2	X	X	X		X
T5	Suction KO drum level HIGH-HIGH	LSHH-4510	> 80 %	1oo1	X		X		X
T6	Suction pressure LOW-LOW	PSLL-4501	< 0.3 barg	1oo1	X	X			X

Legend: X = initiator (row) actuates that effect (column). Voting: 1oo1/1oo2/2oo3 = number of transmitters that must agree to trip.

EFFECT ID	FINAL ELEMENT / ACTION
EFF-1	TRIP COMPRESSOR MOTOR KC-4501
EFF-2	OPEN ANTI-SURGE RECYCLE FV-4502
EFF-3	CLOSE SUCTION XV-4501
EFF-4	MAINTAIN SEAL GAS N2
EFF-5	ANNUNCIATE ALARM ON DCS

#	START PERMISSIVE (AND-gate)	SIGNAL
1	Lube oil pressure OK (> 2 barg)	PSLL-4504
2	Seal gas established	FT-4506
3	KO drum level normal (< 50%)	LT-4510
4	Jacket water flow OK	FSL-4508
5	No active trip / reset done	DCS reset
=>	ALL permissives TRUE (AND) => START KC-4501 ENABLED	RUN

NOTES: 1. Trip set points are DUMMY training values. 2. On any trip the effects marked X are actuated by the Safety PLC and annunciated on the DCS. 3. A trip is latched and requires a manual reset once the cause has cleared and permissives are healthy. 4. Refer to P&ID TJC-LLD-PID-4501 for instrument loops and the Datasheet for design limits of KC-4501.